

Post Composer Mini-Project Setup Guide

Linux Environment Configuration & Execution Guide (Ubuntu, Debian, Fedora, Arch)

This document provides complete instructions to set up, configure, and execute the **Post Composer** mini-project on Linux distributions. The project is composed of a **Spring Boot** backend (utilizing an in-memory H2 database) and a **React + Vite** frontend.

Prerequisites Installation

Ensure the required runtimes (JDK 17/21 and Node.js) are installed on your Linux machine before running the application.

1. Java Development Kit (JDK 17 or JDK 21)

Check if Java is already installed with `java -version`. If not, install via your package manager:

```
# Ubuntu / Debian / Mint:
sudo apt update && sudo apt install -y openjdk-21-jdk

# Fedora / RHEL:
sudo dnf install -y java-21-openjdk-devel

# Arch Linux:
sudo pacman -S jdk21-openjdk
```

2. Node.js & npm (Node v18+ recommended)

Verify installation with `node -v` and `npm -v`. To install:

```
# Ubuntu / Debian (NodeSource LTS):
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs

# Fedora / Arch Linux:
sudo dnf install -y nodejs npm # Fedora
sudo pacman -S nodejs npm # Arch Linux
```

1. Running the Backend (Spring Boot)

The backend uses Spring Boot with an in-memory H2 database. No standalone database server (e.g. MySQL/PostgreSQL) is required.

```
# Step 1: Open a terminal and navigate to the backend directory
cd Experiment-5/backend

# Step 2: Grant execute permissions to the Gradle wrapper
chmod +x gradlew

# Step 3: Launch the Spring Boot application
./gradlew bootRun
```

- Server runs on **http://localhost:8080**.
- H2 Console is accessible at **http://localhost:8080/h2-console** (JDBC URL: jdbc:h2:mem:testdb).
- **Keep this terminal window running.**

2. Running the Frontend (React + Vite)

The client application is built with React and Vite.

```
# Step 1: Open a NEW terminal tab/window and navigate to the frontend directory
cd Experiment-5/frontend

# Step 2: Install project dependencies (first time only)
npm install

# Step 3: Start the development server
npm run dev
```

- The Vite development server will be hosted at **http://localhost:5173**.
- Open your web browser (Firefox, Google Chrome) and navigate to **http://localhost:5173**.

Testing the Application

1. Open **http://localhost:5173** in your browser.
2. Select a target social platform (**Twitter**, **Instagram**, or **Facebook**).
3. Type into the composer and observe the real-time word / character counter.
4. If you exceed the platform limit, a red global error banner will appear.
5. Click **Post** to store the message. It will immediately show in **Recent Posts**, where you can edit or delete entries.

Linux Troubleshooting & Quick Fixes

```
# 1. Port 8080 or 5173 already occupied:
sudo lsof -i :8080 # Find PID using port 8080
kill -9 <PID> # Terminate process

# 2. Permission denied on Gradle wrapper:
chmod +x gradlew

# 3. Clean and rebuild backend if build cache fails:
./gradlew clean bootRun

# 4. Reinstall Node dependencies if npm packages get corrupted:
rm -rf node_modules package-lock.json && npm install
```